

Answer **all** questions.

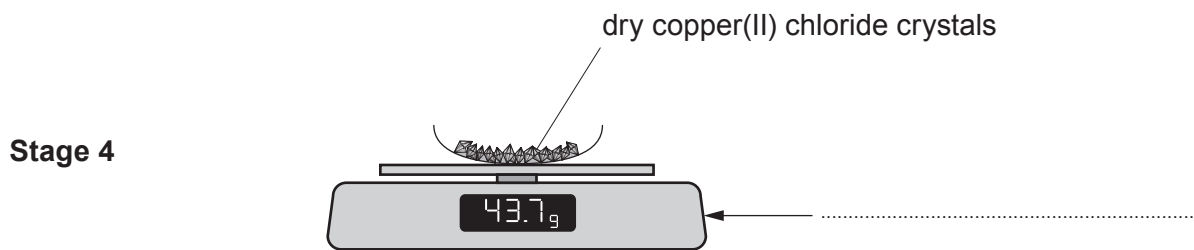
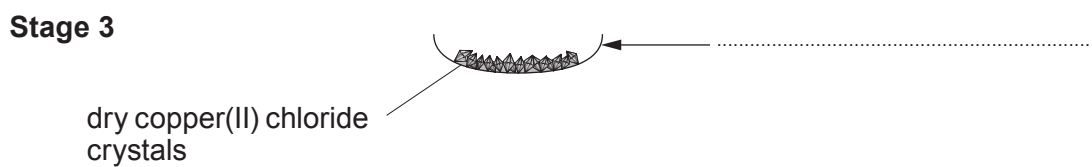
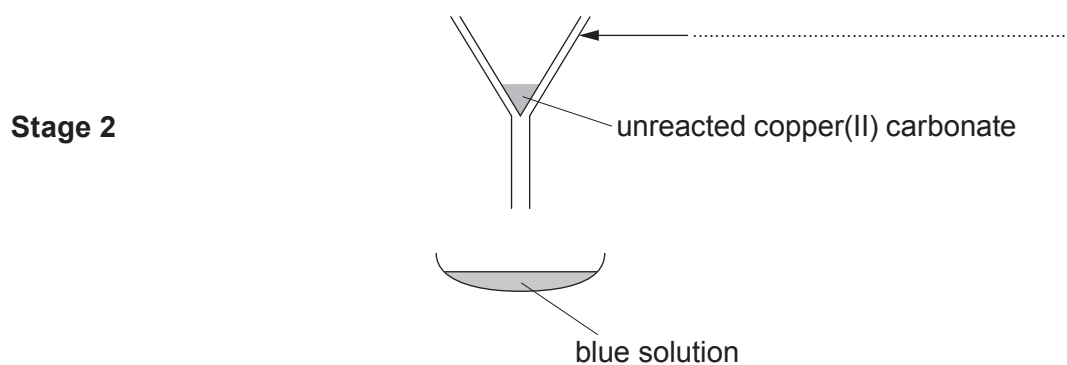
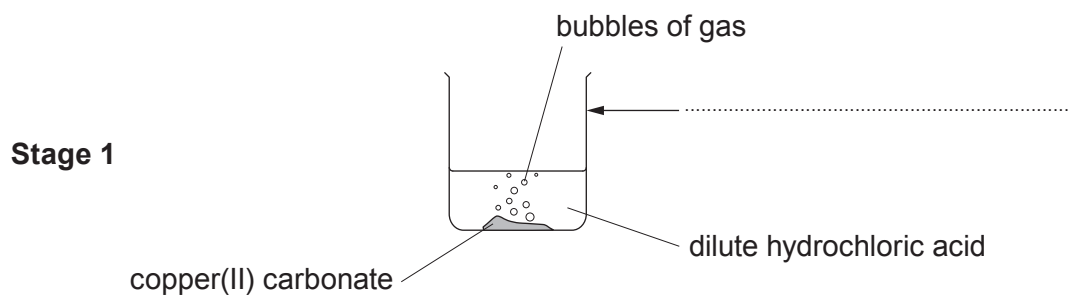
1. Selwyn carried out an experiment to prepare copper(II) chloride crystals.

The diagrams show the stages of the experiment he carried out.

- (a) Choose apparatus from the box to label the diagrams.

[4]

evaporating basin	conical flask	filter funnel	electronic balance
filter paper	beaker	test tube	



- (b) In **Stage 1** Selwyn added copper(II) carbonate until **all** the dilute hydrochloric acid was used up.

Tick (✓) the box next to the statement which best describes what Selwyn would see when all the acid had been used up. [1]

bubbling increases

☐

bubbling stops

☐

bubbling decreases

☐

- (c) The gas formed in **Stage 1** turns limewater milky. Underline the name of this gas. [1]

oxygen

hydrogen

carbon dioxide

nitrogen

- (d) Choose words from the box to complete the following sentences. [2]

evaporation

filtration

distillation

neutralisation

The process used to remove the unreacted copper(II) carbonate in **Stage 2** is called

.....

The process used to remove water in **Stage 3** is called

- (e) The container holding the crystals has a mass of 29.8 g. Using the information given in **Stage 4**, calculate the mass of the crystals formed. [1]

Mass of crystals formed = g

